

Writing Functions in Java

Practice set 1 — declaring functions with inputs and outputs | 10 questions

How to attempt these questions

Each question describes a small real-world job. It tells you what information comes in, what has to be done with it, and what has to be found out or produced at the end. Your job is to turn that description into one Java function.

- Choose a clear name for the function that says what it does.
- Decide what the function takes in: how many things, and what kind each one is (whole number, number with decimals, text, yes/no, a list, or a bundle of related information).
- Decide what the function gives back: one value, several values packed together, a whole record, or nothing at all.
- Write the body so the function does the job, then write one or two lines that call it with the sample values given and print the result.

Some questions take one thing in, some take several, one takes nothing at all. Some give back a number, some a yes/no, some a letter or a line of text, one gives back two numbers together, one gives back a complete record, and one gives back nothing. Read carefully: the shape of the inputs and the output is what each question is really testing.

Question 1: Can this person vote?

One piece of information in, a yes/no decision out

The election office in your ward is checking new applicants for the voter list. The rule is simple: a person must be 18 years old or above to be added.

Information given	The age of the applicant, in completed years (a whole number such as 17, 18, 45).
What to do	Check the age against the rule.
What to find	Whether the applicant is eligible to vote: a plain yes or no.

If the information given is	the answer should be
Age 17	No
Age 18	Yes
Age 62	Yes

Question 2: Monthly electricity bill

Two pieces of information in, one amount out

The electricity board sends a monthly bill to every household. The bill is worked out from how much electricity was used and the price charged for each unit.

Information given	The number of units used in the month (a whole number, for example 240), and the price of one unit in rupees (a value with decimals, for example 6.75).
What to do	Multiply the units used by the price per unit.
What to find	The total amount the household has to pay, in rupees (may have paise, so decimals are allowed).

If the information given is	the answer should be
240 units, Rs 6.75 per unit	Rs 1620.00
100 units, Rs 5.50 per unit	Rs 550.00
0 units, Rs 6.75 per unit	Rs 0.00

Question 3: Grade letter for a student

One number in, one letter out

A college prints a single letter grade on every mark sheet based on the student's overall percentage. The college uses this scale: 90 and above is A, 75 to below 90 is B, 60 to below 75 is C, 40 to below 60 is D, and anything below 40 is F.

Information given	The student's overall percentage (a value with decimals, between 0 and 100, for example 78.5).
What to do	Find which band on the scale the percentage falls into.
What to find	A single letter: A, B, C, D or F.

If the information given is	the answer should be
92.0	A
78.5	B
39.9	F

Question 4: Is this a valid Indian mobile number?

Text in, a yes/no decision out

A customer-care desk collects mobile numbers from walk-in customers on a paper form and later types them into the system. Many entries are typed wrongly. A number is accepted only if it has exactly 10 characters, every character is a digit, and the first digit is 6, 7, 8 or 9.

Information given	The mobile number exactly as typed, as text (for example "9845012345" or "9845012345").
What to do	Check the length, check that every character is a digit, and check the first digit.
What to find	Whether the number is acceptable: yes or no.

If the information given is	the answer should be
"9845012345"	Yes
"5845012345"	No (starts with 5)
"98450 12345"	No (has a space, 11 characters)
"984501234"	No (only 9 characters)

Question 5: City bus fare

A number and a yes/no in, an amount out

The city bus service charges a flat Rs 10 for boarding plus Rs 2 for every kilometre travelled. Senior citizens get a 50% discount on the whole fare.

Information given	The distance travelled in kilometres (a value with decimals, for example 7.5), and whether or not the passenger is a senior citizen (a simple yes/no).
What to do	Work out the normal fare, then halve it if the passenger is a senior citizen.
What to find	The fare to be charged, in rupees (decimals allowed).

If the information given is	the answer should be
7.5 km, not a senior citizen	Rs 25.00
7.5 km, senior citizen	Rs 12.50
0 km, not a senior citizen	Rs 10.00

Question 6: Print a shop receipt line

Several pieces of information in, nothing given back (only printing)

A small kirana shop wants a receipt line printed on the screen for each item a customer buys. The line should look like this:

Ravi | Sugar 1kg | 2 x Rs 48.00 = Rs 96.00

Information given	The customer's name (text), the item name (text), the quantity bought (a whole number), and the price of one unit in rupees (decimals allowed).
What to do	Multiply quantity by unit price to get the line total, then print the line in the format shown.
What to find	Nothing is handed back to the caller. The only result is the printed line on the screen.

If the information given is	the answer should be
Ravi, Sugar 1kg, 2, 48.00	prints: Ravi Sugar 1kg 2 x Rs 48.00 = Rs 96.00
Meena, Milk 500ml, 3, 27.50	prints: Meena Milk 500ml 3 x Rs 27.50 = Rs 82.50

Question 7: Busiest and quietest day at the public library

A list of numbers in, two numbers out

The city public library counts how many visitors came in each day. At the end of the month the librarian wants to know both the highest daily count and the lowest daily count from that month.

Information given	A list of daily visitor counts for the month (whole numbers, one for each day, at least one day).
What to do	Go through the list once and keep track of the largest and the smallest value seen.
What to find	Two whole numbers together: the highest count and the lowest count.

If the information given is	the answer should be
[120, 95, 210, 180, 60]	highest 210, lowest 60
[75]	highest 75, lowest 75

Question 8: Address label for a citizen

One bundle of related information in, one line of text out

The ward office posts property-tax notices to residents. Each resident's details are kept together as one record: name, house number, street name, ward number and PIN code. The office needs a single line of text for the address label, in this shape:

Lakshmi Rao, #42, 5th Cross, Ward 31, Bengaluru - 560035

Information given	One citizen record that holds five things: name (text), house number (text, because it may be like "42A"), street (text), ward number (whole number), PIN code (whole number).
What to do	Join the five pieces of information, with the fixed words and punctuation shown, into one line.
What to find	The finished address label as a single line of text.

If the information given is	the answer should be
Lakshmi Rao, 42, 5th Cross, 31, 560035	"Lakshmi Rao, #42, 5th Cross, Ward 31, Bengaluru - 560035"
Imran K, 7B, Main Road, 31, 560102	"Imran K, #7B, Main Road, Ward 31, Bengaluru - 560102"

Question 9: Register a new customer

Three pieces of information in, one complete record out

A gym at the front desk registers new members. The staff enter the person's name, email and mobile number. The system must create a complete member record that also carries a membership number. Membership numbers start at 1001 and go up by one for each new member registered.

Information given	The member's name (text), email (text) and mobile number (text).
What to do	Give the next membership number and put all four things together into a single member record.
What to find	The complete member record (membership number, name, email, mobile) handed back as one unit.

If the information given is	the answer should be
"Anjali", "anjali@mail.com", "9900011122"	record: 1001, Anjali, anjali@mail.com, 9900011122
next call: "Kiran", "kiran@mail.com", "9900033344"	record: 1002, Kiran, kiran@mail.com, 9900033344

Question 10: Standard water tariff

Nothing in, one fixed value out

The water supply board charges a standard rate per kilolitre of water for households. Several parts of the billing program need this rate, and the board wants it defined in exactly one place so that when the rate changes, it is changed once. The current rate is Rs 7.00 per kilolitre.

Information given Nothing. The rate is decided by the board and does not depend on any input.

What to do Simply hand back the current standard rate.

What to find The rate in rupees per kilolitre (decimals allowed).

If the information given is	the answer should be
(no input)	7.00